

Bangladesh University of Business and Technology (BUBT)



LAB Report

Course Code :
Course Title :
Experiment No. :
Experiment Name :

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Intake : 53
Section : 01
Program : B.Sc. in CSE

Name:

Dept. of
Bangladesh University of
Business and Technology

Date of Submission :

Signature of Teacher

Experiment Name: Wireless Routing Protocol

Introduction:

Wireless Routing allows devices to connect and communicate without cables using a wireless router. In this experiment, DHCP (Dynamic Host Configuration Protocol) is used to automatically assign IP addresses to devices, making network configuration easier and faster. It is commonly used in homes and offices.

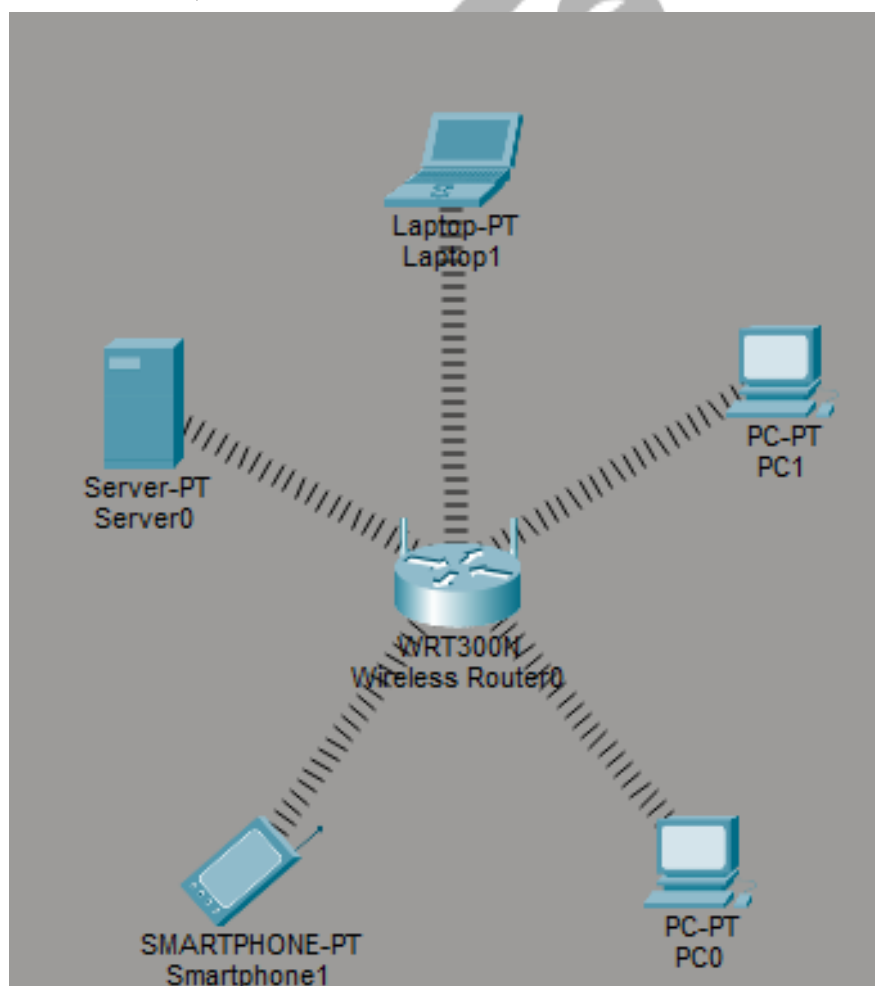
Software and Tools Used:

1. Cisco Packet Tracer (Network Simulation Software)
2. Wireless Router / Access Point
3. PCs / Laptops / Smartphones (Wireless Devices)
4. Switch (if required)
5. Ethernet Cables
6. Config Tab Interface

All devices were simulated virtually inside Cisco Packet Tracer software.

Procedure:

1. Open Cisco Packet Tracer and place 2 PCs, 1 Laptop, 1 Smartphone, 1 Server, and a wireless router.



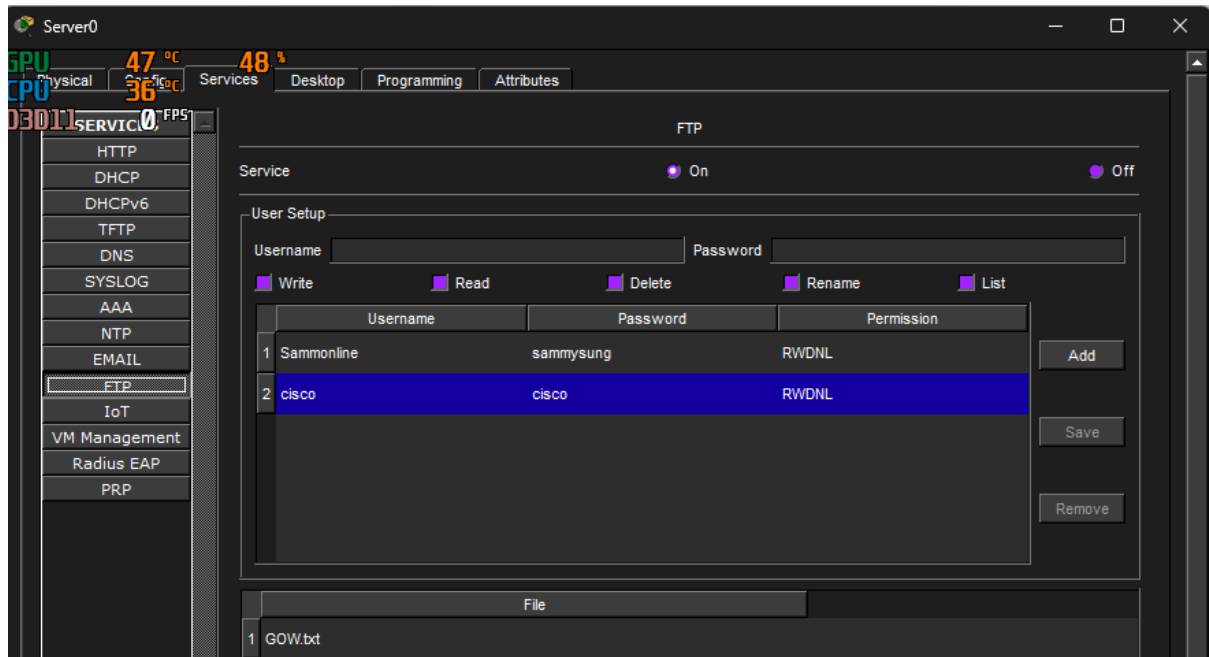
2. Configure the wireless router in GUI by setting SSID, security type, password, and enabling DHCP.

The screenshot shows the 'Wireless Router0' GUI. The 'Setup' tab is active, displaying various configuration sections. The 'Internet Setup' section includes 'Internet Connection type' set to 'Automatic Configuration - DHCP', 'Host Name', 'Domain Name', and 'MTU' (1500). The 'Network Setup' section includes 'Router IP' (192.168.10.1) and 'Subnet Mask' (255.255.255.0). The 'DHCP Server Settings' section shows 'DHCP Server' as 'Enabled', 'Start IP Address' (192.168.10.11), 'Maximum number of Users' (10), and 'IP Address Range' (192.168.10.11 - 20). A 'Help...' button is visible on the right.

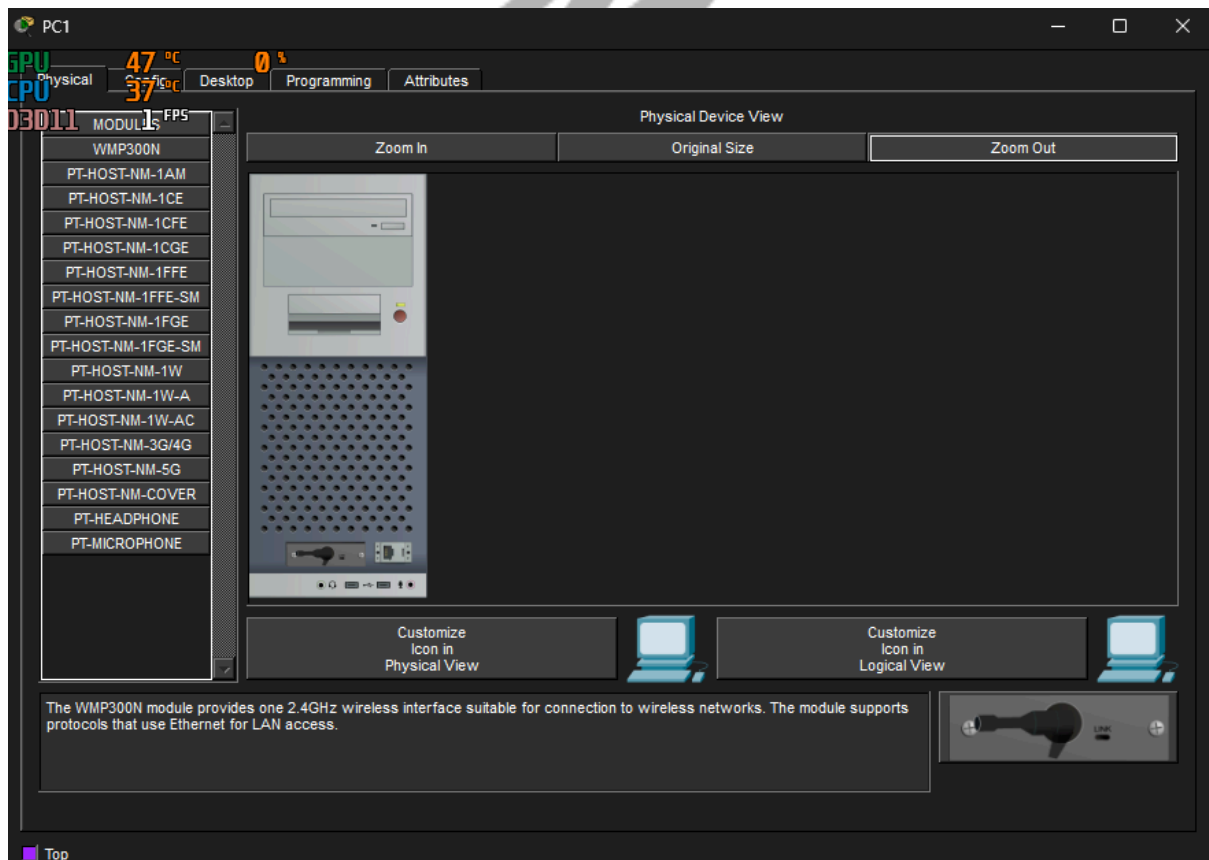
This section of the GUI shows wireless settings. 'Network Mode' is set to 'Mixed'. 'Network Name (SSID)' is 'Dot Net'. 'Radio Band' is 'Standard - 20MHz Channel'. 'Wide Channel' is '3'. 'Standard Channel' is '1 - 2.412GHz'. 'SSID Broadcast' is 'Enabled'.

This section shows security settings. 'Security Mode' is 'WPA Personal'. 'Encryption' is 'AES'. 'Passphrase' is 'Ca#30032026'. 'Key Renewal' is '3600 seconds'.

3. Configure the server and enable FTP service with username "Sammonline" and password "sammysung".

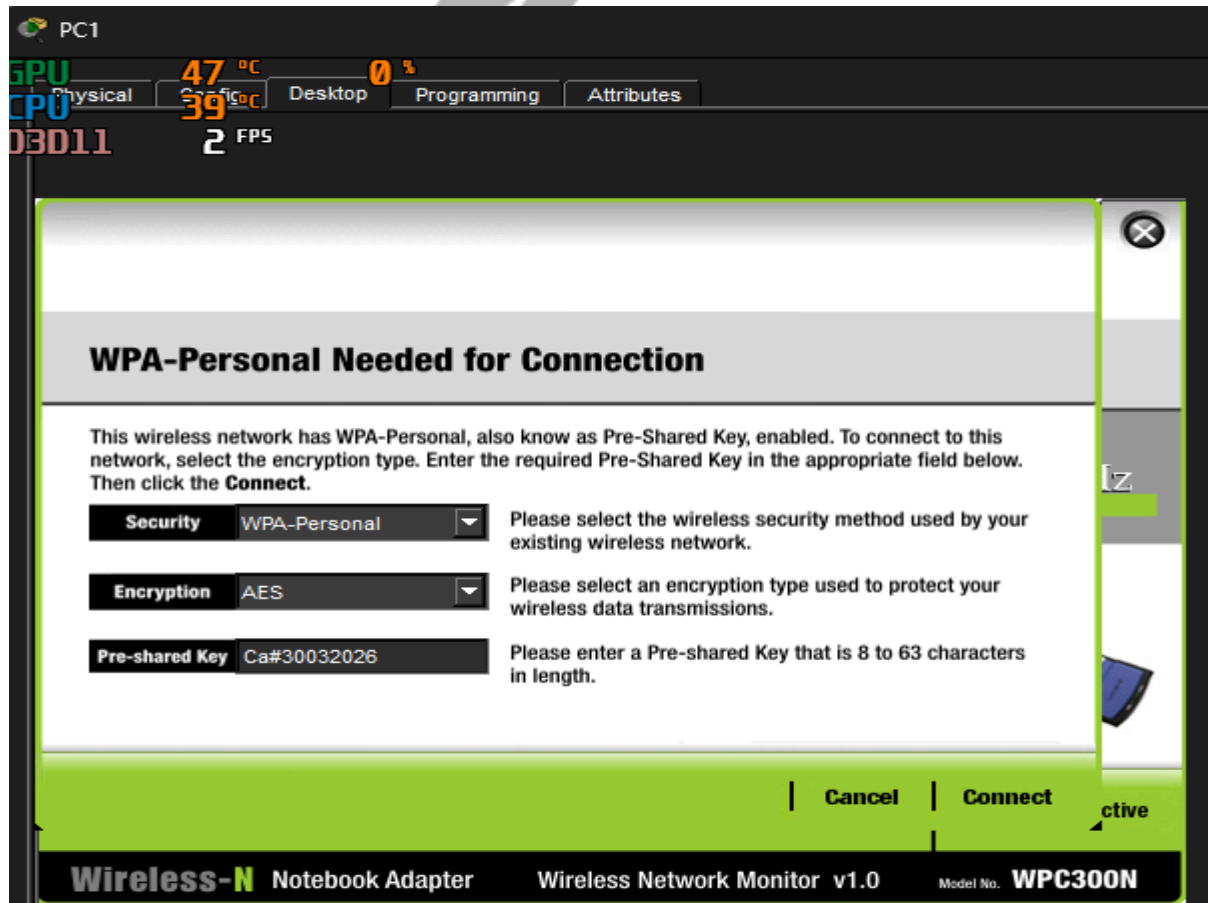


- Convert one wired PC and the server to wireless by turning them off, adding a wireless module (WMP300N) from the Physical tab, and turning them back on.

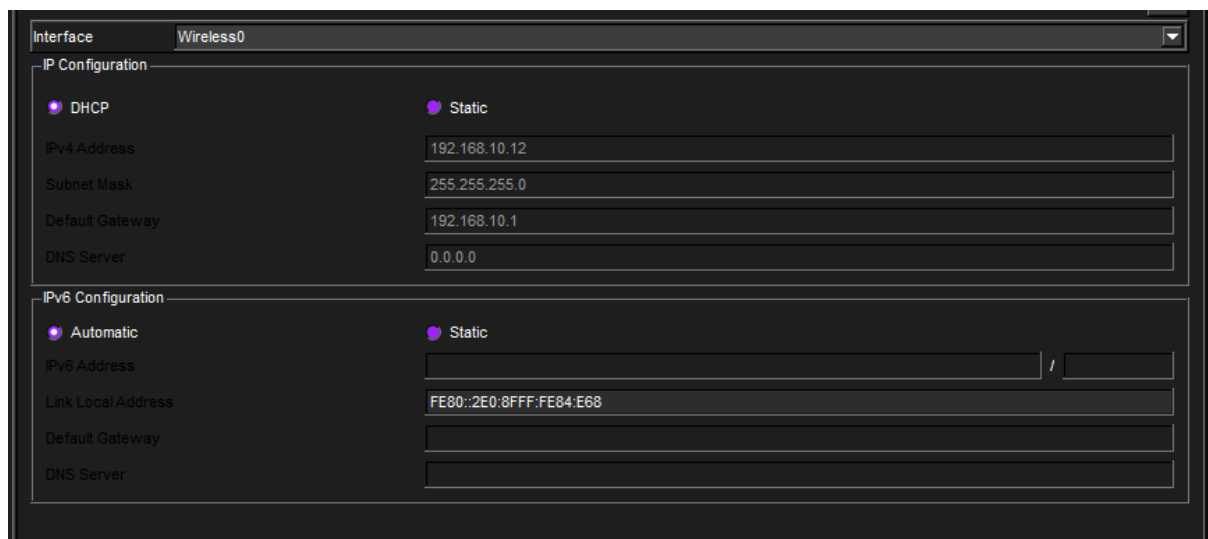




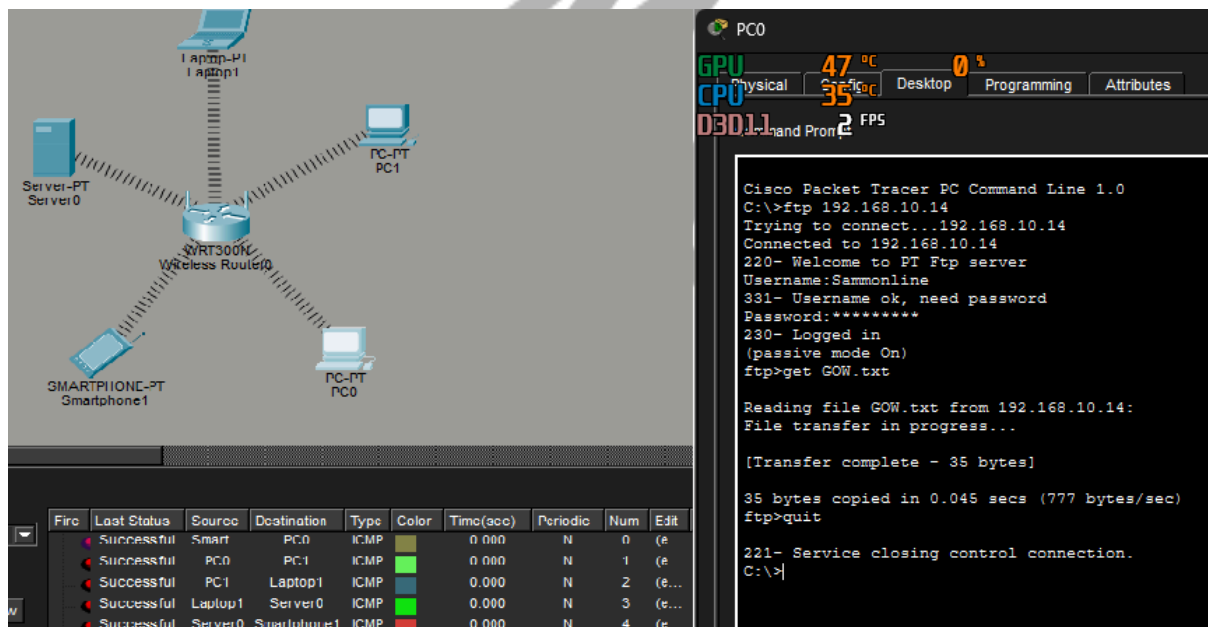
5. Connect all wireless devices (PCs, laptop, smartphone, and server) to the Wi-Fi network using SSID and password.



6. Ensure all devices obtain IP addresses automatically using DHCP (configured in GUI).



7. Test connectivity by sending ping between devices and access the FTP server using given credentials.



Conclusion:

In this experiment, we successfully configured a wireless network using DHCP. Devices automatically received IP addresses and communicated without manual configuration. This method is simple, efficient, and widely used in real-world networks.